$T: \mathcal{H} \to \mathcal{H}$
$\psi \to T(\psi)$ leaves transition amplitudes invariant

if T α linear, unitary operator

→ consequence of fact that $\psi \sim \lambda\psi$ $\forall \lambda \neq 0$

physical states live in $P\mathcal{H}$ (projective)

<u>Aside:</u> global $U(1)$ transformation $\psi(x) \to e^{i\theta}\psi(x)$ θ fixed

local $U(1)$ transformation $\psi(x) \to e^{i\theta(x)}\psi(x)$ θ function of x

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assume that $\mathcal{H}$ finite-dimensional.

Want to know the action of a group of symmetries G, then we must have linear, <u>proportional</u> to unitary action

→ looking at projective representations

- $GL(\mathcal{H})$ = invertible linear maps from $\mathcal{H} \to \mathcal{H}$

  regular rep: $\rho: G \to GL(\mathcal{H})$ smooth and $\rho(gh) = \rho(g)\rho(h)$

- $PGL(\mathcal{H}) = GL(\mathcal{H}) / \{\lambda \mathbb{1} \mid \lambda \neq 0\}$ ← still a group

  projective rep: $\rho: G \to PGL(\mathcal{H})$

if $\rho: G \to GL(\mathcal{H})$ is a rep, $\tilde{\rho}: G \to PGL(\mathcal{H})$ is a proj. rep.
$g \mapsto [\rho(g)]$

<u>converse is not true</u> → means that proj. rep. are more general

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if G is simply connected, then any proj rep. can be lifted to a regular rep. (for finite-dim. rep)

<u>lift</u> (choose rep. from each eq class)

$$\rho(gh) = c(g,h)\,\rho(g)\,\rho(h) \qquad \left([\rho(gh)] = [\rho(g)]\,[\rho(h)] \right)$$

<u>Heisenberg algebra</u> $[q,p] = z \qquad [z,q] = [z,p] = 0$

$$z|\psi\rangle = -i\hbar|\psi\rangle \quad \forall |\psi\rangle$$